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Hierarchical Vision Transformers for prostate biopsy grading: Towards bridging the generalization gap

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ABSTRACT

Practical deployment of Vision Transformers in computational pathology has largely been constrained by the sheer size of whole-slide images. Transformers faced a similar limitation when applied to long documents, and Hierarchical Transformers were introduced to circumvent it. This work explores the capabilities of Hierarchical Vision Transformers for prostate cancer grading in WSIs and presents a novel technique to combine attention scores smartly across hierarchical transformers. Our best-performing model matches state-of-the-art algorithms with a 0.916 quadratic kappa on the Prostate cANcer graDe Assessment (PANDA) test set. It exhibits superior generalization capacities when evaluated in more diverse clinical settings, achieving a quadratic kappa of 0.877, outperforming existing solutions. These results demonstrate our approach's robustness and practical applicability, paving the way for its broader adoption in computational pathology and possibly other medical imaging tasks. Our code is publicly available at https://github.com/computationalpathologygroup/hvit.

1. Introduction

Prostate cancer is the second most frequently diagnosed cancer and the fifth leading cause of cancer death among men globally. Prostate biopsy grading by pathologists plays a critical role in determining the aggressiveness of prostate cancer, thereby guiding interventions ranging from active surveillance to surgical procedures. With the rising number of prostate cancer patients placing increasing pressure on pathologists, there is a pressing need for computational methods to assist in daily workflows.

Digitization of histological slides has contributed to the growing availability of large datasets, fostering computer vision research opportunities to support and augment pathologists with deep learning algorithms. Deep learning has revolutionized various domains of computer vision, demonstrating unprecedented successes in tasks like image classification, object detection, and semantic segmentation. While convolutional neural networks (CNNs) marked significant progress a decade ago (Krizhevsky et al., 2012), recent attention-based models like the Vision Transformer (ViT) (Dosovitskiy et al., 2021) have further pushed the boundaries of performance.

However, computational pathology presents a unique set of challenges due to the enormous sizes of whole-slide images (WSIs), which exceed the memory capacities of conventional deep learning hardware. Consequently, innovative strategies have emerged to overcome this memory bottleneck. A prevailing approach involves partitioning these massive images into smaller patches. These patches often serve as the input units, with labels derived from pixel-level annotations (Ehteshami Bejnordi et al., 2017; Coudray et al., 2018). Yet, obtaining pixel-level annotations is labor-intensive and sometimes impractical, especially in complex tasks like prostate cancer grading where pathologists can only annotate what they recognize. Therefore, recent research has explored techniques beyond full supervision, focusing on more flexible training paradigms, such as weakly supervised learning.

Weakly supervised learning exploits coarse-grained (image-level) information to infer fine-grained (patch-level) details automatically. Multiple Instance Learning (MIL) has recently emerged as a powerful, weakly-supervised approach in several computational pathology challenges, where it has shown remarkable performance (Hou et al.,

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2015; Campanella et al., 2019). By enabling the analysis of whole-slide images with slide-level labels only, it has circumvented the need for pixel-level annotations, facilitating tasks such as risk prediction and genetic mutation detection (Schmauch et al., 2020; Garberis et al., 2022).

Despite their success, most MIL methods neglect the spatial relationships among patches, overlooking valuable contextual information. To address this limitation, research focused on developing methods capable of integrating broader context (Lerousseau et al., 2021; Pinckaers et al., 2021; Shao et al., 2021). Among these, Hierarchical Vision Transformers (H-ViTs) have emerged as a promising solution, achieving state-of-the-art results in cancer subtyping and survival prediction (Chen et al., 2022).

Motivated by these considerations, we present a comprehensive analysis of H-ViTs in the context of prostate cancer grading. Our work makes several pivotal advancements in the field, specifically:

1. We show H-ViTs match state-of-the-art prostate cancer grading algorithms when tested on cases from the same center as the training data, while demonstrating stronger generalization to more diverse clinical settings.
2. We demonstrate the benefit of prostate-specific pretraining over more generic multi-organ pretraining.
3. We systematically compare two H-ViT variants and offer concrete guidance on when each is preferable.
4. We provide an extensive analysis of loss function choices for ordinal classification, showcasing the superiority of treating prostate cancer grading as a regression task.
5. We enhance model interpretability by introducing an innovative approach for combining attention scores across hierarchical transformers which balances task-agnostic and task-specific contributions.

2. Related work

2.1. Weakly supervised learning

Campanella et al. (2019) demonstrated that when provided with sufficient data, weakly supervised models can perform comparably to fully supervised methods trained using pixel-level annotations. They model the whole-slide image as a bag of instances and leverage the standard MI assumption: positive bags must contain at least one positive instance. In contrast, negative bags exclusively consist of negative instances. During training, the instance with highest likelihood within a bag is selected, and the loss is computed using the whole-slide label as the target. One limitation of this modeling is that it lets a single instance decide the label of the entire slide. To address this, later developed MIL-based methods harness pre-trained encoders for instance-level feature extraction and train a lightweight aggregation network to optimally combine these features into a single slide-level prediction (Tellez et al., 2020, 2021; Zhang et al., 2022). Lu et al. (2021) further introduced an attention-gating mechanism that provides a form of interpretability.

Despite their success, patch-based methods are inherently limited. They rely on the implicit assumption that the signal is local, as it is bounded by the size of the patches fed into the network. As a result, they fall short when the information is on a bigger scale or require broader context. Methods capable of integrating more context were developed to address this limitation.

2.2. Integrating spatial context

Pinckaers et al. (2021) developed Streaming, which enables end-to-end training on large inputs by exploiting the inherent locality of most transformations in convolutional neural networks. They achieve this by breaking a whole slide image into smaller images sequentially processed until the stitched output fits within GPU memory. Despite handling entire slides, Streaming suffers from additional processing time and memory usage due to the temporary storage of intermediate forward and backward maps. Lerousseau et al. (2021) introduced SparseConvMIL, a context-aware MIL approach that builds a sparse map by associating patch embeddings with their original position in the WSI. A sparse-input CNN then processes this sparse map to generate an embedding for the WSI, subsequently classified using a generic classifier. It has demonstrated improved performance for pan-cancer subtype classification.

Computational pathology has also witnessed a growing interest in Transformers (Vaswani et al., 2017), which better capture global context than CNNs due to their ability to model correlations and distant relationships among instances in a sequence. Shao et al. (2021) developed a novel correlated MIL framework from which they derive a Transformer-based architecture, which shows great potential in applications such as survival prediction and cancer cell spread detection. Coupled with tailored self-supervised learning techniques such as DINO (Caron et al., 2021), Vision Transformers (ViTs) have fostered substantial progress in computer vision by enabling models to learn meaningful representations from unlabeled data.

2.3. Hierarchical vision transformers

Standard ViTs model images as sequences of fixed-size patches, which may not efficiently capture multi-scale contextual information, especially in gigapixel images like WSIs. To address this limitation, Hierarchical Vision Transformers have been proposed, introducing a multi-scale, hierarchical architecture that allows for the integration of features at various levels of granularity. Architectures such as Swin Transformer (Liu et al., 2021), ViL (Zhang et al., 2021), and Pyramid Vision Transformer (Wang et al., 2021) have demonstrated the effectiveness of hierarchical designs in capturing both local and global contextual information in natural images.

In computational pathology, Chen et al. (2022) pioneered Hierarchical ViTs by exploiting the multi-scale structure of whole slide images to design a three-stage architecture. They adapted the DINO framework to pretrain the first two ViTs with unlabeled data, enabling efficient learning of context-aware image representations and achieving state-of-the-art performance in cancer subtyping and survival prediction.

2.4. Generalizability of prostate cancer grading models

Prostate biopsy grading models have demonstrated significant potential in various studies (Bulten et al., 2020; Nagpal et al., 2020; Strom et al., 2020). Despite claims of robust generalization to external datasets, most of these studies evaluate models on homogeneous data, making it difficult to assess their performance in real-world clinical settings. Bulten et al. (2022) shows that the top-performing models from the PANDA challenge not only matched but often surpassed the performance of previous works when evaluated on internal validation sets. Yet, when tested on external sets showing more diversity, their performance tends to decline. For example, the average quadratic weighted kappa among the best PANDA challenge models was 0.937 on the internal validation set but dropped to 0.862 and 0.868 on two external validation sets.

To better assess generalization, Faryna et al. (2024) proposed evaluating algorithms on a broader range of cases from multiple institutions, simulating real-world conditions. They curated a diverse dataset of prostate biopsies through crowdsourcing. Only two of the five top-ranked public AI algorithms they evaluated show decent generalization, with the best model achieving a quadratic weighted kappa of 0.862. These consistent drops in performance underscore an ongoing challenge in achieving true generalizability that current methodologies have yet to fully overcome.

2.5. Model interpretability in computational pathology

In computational pathology, the interpretability of deep learning models is crucial for gaining the trust of clinicians and facilitating integration into clinical workflows. It primarily involves identifying the regions that contribute most to the model's prediction. In clinical practice, these regions can be compared with expert assessments to validate the method's reliability. In research settings, highlighting morphological features that characterize important regions can aid in biomarker discovery (L'Imperio et al., 2023). However, due to the inherent complexity of deep learning models, pinpointing these regions and fully understanding the decision-making process remains a challenging task.

To address this challenge, feature attribution methods like gradient-based importance techniques have been proposed (Selvaraju et al., 2016; Chattopadhay et al., 2018; Bousselham et al., 2024). The idea is to trace back the most activated features to the input in order to generate a saliency map where each element of the input is assigned a score reflecting its contribution to the model's prediction. Alternatively, attention-based methods can directly translate the learned attention weights into saliency maps for explaining the prediction. Popularized by Lu et al. (2021), attention heatmaps have since become the default visualization method for model interpretability in attention-based computational pathology models (Shao et al., 2021; Dooper et al., 2023).

In H-ViTs, features are processed through several transformer blocks at different hierarchical levels, making it non-trivial to aggregate attention scores from each level into a coherent interpretability map. Chen et al. (2022) attempted to address this by averaging attention scores from multiple levels into hierarchical attention maps. However, their approach is limited as they do not derive attention scores for all transformers and rely on simple averaging, which may overlook the complex interactions between hierarchical features. Effectively merging attention scores from multiple consecutive Vision Transformers remains an open problem.

3. Methods

3.1. Hierarchical vision transformer

The inherent hierarchical structure within whole-slide images spans across various scales, from tiny cell-centric regions – (16,16) pixels at 0.50 microns per pixels (mpp) – containing fine-grained information, all the way up to the entire slide, which exhibits the overall intra-tumoral heterogeneity of the tissue microenvironment. Along this spectrum, (256,256) patches depict cell-to-cell interactions, while larger regions – (1024,1024) to (4096,4096) pixels – capture macro-scale interactions between clusters of cells.

Drawing inspiration from this layered structure, Chen et al. (2022) introduced the Hierarchical Image Pyramid Transformer (HIPT), a three-stage architecture built upon H-ViTs. First, a Vision Transformer ($\text{ViT}_{256}-16$, referred to as patch-level Transformer in what follows) performs cell-level aggregation by breaking down 256×256 patches into 16×16 tokens. Then, another Transformer ($\text{ViT}_{4096}-256$, referred to as region-level Transformer in what follows) builds context-aware embeddings by aggregating non-overlapping 256×256 patches within larger 4096×4096 regions. Finally, a last Transformer ($\text{ViT}_{\text{WSI}}-4096$, referred to as slide-level Transformer in what follows) pools the resulting region-level tokens into a single slide-level representation that is projected to class logits for loss computation.

In this work, we experiment with two variants of the previously described architecture. The first, named Global H-ViT, replicates the best configuration presented in Chen et al. (2022): both patch-level and region-level Transformers are pretrained and kept frozen. Only the slide-level Transformer is trained on the downstream task. The second variant, deemed Local H-ViT, consists of pretrained patch-level and region-level Transformers, with only the patch-level Transformer kept frozen. The region-level and slide-level Transformers are jointly trained on the downstream task (Fig. 1).

The lack of independent testing in Chen et al. (2022) raises questions about the generalizability of their conclusions, especially in a more complex setting. Grading prostate cancer from biopsies involves complexities not seen in the classification challenges discussed in Chen et al. (2022). We thus rigorously examine four critical aspects to identify the optimal model configuration: self-supervised pretraining, model variant, context size, and the choice of loss function. The potential interdependence among these components adds a layer of complexity to our investigation.

3.2. Self supervised pretraining

Our study explores the benefits of pretraining both patch-level and region-level Transformers exclusively using prostate biopsies from the PANDA dataset. To allow for direct comparison with Chen et al. (2022), we also adopt DINO (Caron et al., 2021) as self-supervised pretraining framework.

Pretraining the patch-level transformer. Non-overlapping (256, 256) patches are extracted from each slide in the training set at 0.50 mpp. Patches with less than 10% tissue are discarded, resulting in roughly 2.4M[1] patches. These are then used as inputs to pretrain a ViT-Small (ViT-S) with DINO. DINO is a knowledge distillation framework where a student network g_{θ_s} is trained to match the output of a teacher network g_{θ_t}, parameterized by θ_s and θ_t respectively. Local and global crops are applied as data augmentation to encourage local-to-global correspondences between the student and the teacher. From a given image $\mathbf{x}$, a set V of different views is generated: 8 local (96, 96) views and 2 global (224, 224) views. All views are passed through the student, while only the global views are passed through the teacher. Given a pair of input views, both networks output probability distributions over K dimensions denoted by p_s and p_t. These distributions are matched by minimizing the cross-entropy loss with respect to the student network parameters θ_s:

$$\min_{\theta_s} \sum_{\mathbf{v} \in \left\{\mathbf{v}_1^g, \mathbf{v}_2^g\right\}} \sum_{\substack{\mathbf{v}' \in V \\ \mathbf{v}' \neq \mathbf{v}}} H\left(p_t(\mathbf{v}), p_s\left(\mathbf{v}'\right)\right) \tag{1}$$

where $H(a, b) = -a \log b$.

Chen et al. (2022) rightly point out that DINO data augmentation strategy is particularly suitable for histology slides, as local (96, 96) crops at 0.50 mpp capture the context of multiple cells and their surrounding extracellular matrices. Therefore, we stick to DINO default data augmentation: horizontal flips and color jittering are applied to all views, with solarizing used on one of the global views.

To monitor pretraining, we implement a k-nearest neighbors (kNN) evaluation following the methodology described by Caron et al. (2021). To that end, we curate a binary classification dataset with patch-level labels by leveraging the pixel-level annotation masks provided for a subset of the PANDA dataset. We extract benign and tumor[2] patches of size (256, 256) at 0.50 mpp. We only keep tumor patches with more than 10% tissue and where the annotation mask covers more than 80% of the tissue area. Benign patches are only extracted from benign slides to account for potentially missed tumor regions in cancerous slide annotation masks. This results in 973,001 benign patches and 375,852 tumor patches. To create a balanced binary classification dataset out of these patches and ensure the kNN evaluation step takes minimal computation time compared to model pretraining, we randomly sample 5000 benign and 5000 tumor patches. We subsequently split these

[1] average computed across the 5 cross-validation folds

[2] for Radboud slides, we grouped the various Gleason grade annotations into a single "tumor" category

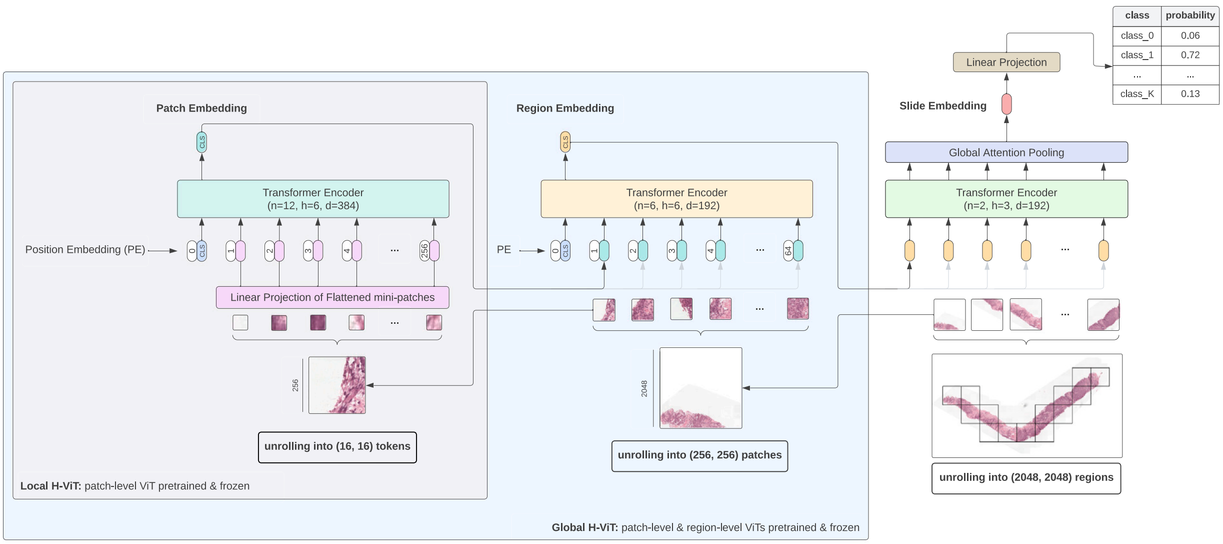


Fig. 1. Overview of our Hierarchical Vision Transformer for whole-slide image analysis. The model processes whole-slide images at multiple scales. Slides are unrolled into non-overlapping 2048 × 2048 regions, which are further divided into 256 × 256 patches following a regular grid. First, a pretrained ViT-S/16 (referred to as the patch-level Transformer) embeds these patches into feature vectors. These patch-level features are then input to a second Transformer (referred to as the region-level Transformer), which aggregates them into region-level embeddings. Finally, a third Transformer (referred to as the slide-level Transformer) pools the region-level embeddings into a slide-level representation, which is projected to class logits for downstream task prediction. We experiment with two model variants: in Global H-ViT, both the patch-level and region-level Transformers are pretrained and frozen, with only the slide-level Transformer undergoing weakly-supervised training; in Local H-ViT, only the patch-level Transformer is frozen, while both the region-level and slide-level Transformers are trained using weak supervision.

patches into train and test sets, with a 60:40 ratio. To prevent data leakage during kNN evaluation, we remove the 4000 test patches from the pretraining dataset.

This evaluation helps tracking student and teacher performance during pretraining and serves as a criterion for early stopping: we pretrain the patch-level Transformer for 50 epochs and stop early when the student outperforms the teacher or when the teacher's performance did not increase for 10 consecutive epochs

Pretraining the region-level transformer. We replicate the hierarchical pretraining stage described in Chen et al. (2022). Non-overlapping (4096, 4096) regions are extracted from each slide in the training set at 0.50 mpp. Regions with less than 10% tissue are discarded, leaving approximately 28k regions per fold. The previously pretrained patch-level Transformer is used as the embedding layer for the region-level Transformer, a ViT-Tiny which also undergoes pretraining via DINO. Each (256, 256) patch in a given (4096, 4096) region is passed through the pretrained patch-level Transformer. The resulting 384-dim [CLS] token is used as embedding for that patch, eventually yielding a (256, 384) dimensional sequence. To enable data augmentations, this sequence is rearranged into a 2D feature grid of shape (16, 16, 384). We perform (6,6), (14,14) local–global crops in matching the scale of (96,96), (224, 224) crops for (256, 256) inputs. Dropout ($p = 0.10$) is applied to all views. We also pretrained the region-level Transformer on smaller-sized regions (Section 5.5).

We do not monitor teacher and student performance throughout pretrainining of the region-level Transformer because we lack reliable region-level labels. Indeed, some data has pixel-level annotations, but going from pixel-level to categorical region-level labels is ambiguous. For example, a region could have multiple Gleason patterns, but there is no pathological definition of combining those to a region-level score.

3.3. Enhanced model interpretability

Our model consists of three hierarchical Transformers, each contributing different attention scores. These can be visualized either individually or combined into more comprehensive factorized heatmaps. This work extends the attention visualization framework of Chen et al. (2022) by offering deeper insights into the patterns identified by the model when trained for specific downstream tasks. We address key limitations of the previous approach and provide a novel method for better understanding the task-specific focus of the model.

Slide-level transformer attention. Chen et al. (2022) visualized attention only for the patch-level and region-level Transformers, which they kept frozen during training. As a result, the generated heatmaps solely show task-agnostic features learnt during self-supervised pretraining. They do not provide any information regarding the relevance of these features for the weakly supervised task at hand, which make these heatmaps impractical. To overcome this limitation, we derive attention heatmaps from the slide-level Transformer, which is explicitly trained for the downstream classification task (Fig. 5(e)). These heatmaps reveal the regions that the model deems important in the context of the weakly supervised learning task, in our case prostate cancer grading. By highlighting task-specific features, these provide more actionable insights and fill the interpretability gap left by the previous method.

Stitched attention heatmaps. Chen et al. (2022) lack a stitched visualization at the whole-slide level. We introduce stitched heatmaps to provide a whole-slide visualization of the model's attention. This comprehensive view offers a more intuitive understanding of which parts of the slide contribute most significantly to the model's decision-making process (Fig. 5). Having stitched attention heatmaps can be particularly useful for clinicians and researchers to pinpoint the critical areas contributing to the model prediction.

Refined attention factorization. Attention scores from the different Transformers hold varying degrees of relevance in highlighting the areas critical in the model's decision process. Indeed, some Transformers are pretrained and frozen, while others are explicitly trained for the downstream classification task. Therefore, we introduce a parameter $\gamma \in [0, 1]$ that controls how attention scores from the different Transformers are combined.

Let N be the total number of Transformers involved in the pretraining or the training processes. Let n be the number of frozen Transformers. For the pixel with (x, y) coordinates in the slide, we denote by $a^i_{(x,y)}$ the attention score of the last self-attention layer of the i-th Transformer T_i. We compute the factorized attention score for that pixel, $a_{(x,y)}$, as:

$$a_{(x,y)} = \frac{1}{\beta} \sum_{i=0}^{N-1} a^i_{(x,y)} \left[\gamma(1 - \mathbb{1}_{\mathrm{F}}(\mathrm{T}_i)) + (1-\gamma)\mathbb{1}_{\mathrm{F}}(\mathrm{T}_i)\right] \quad (2)$$

with $\beta = (N-n)\cdot\gamma + n\cdot(1-\gamma)$ the normalization constant and $\mathbb{1}_{\mathrm{F}}(\mathrm{T}_i)$ the indicator function defined on the set of Transformers $\left\{\mathrm{T}_i\right\}_{i=0}^{N-1}$ and equal to 1 if T_i is frozen, 0 otherwise.

This approach provides a more comprehensive view of the model's decision-making process by enabling adjustments between task-agnostic features learned during self-supervised pretraining and task-specific features derived from weakly-supervised training.

4. Materials

The assessment of Gleason grades provides critical prognostic insights for patients with prostate cancer, ultimately playing a crucial role in determining the course of treatment. Pathologists classify tumors into different growth patterns by analyzing the histological architecture of the tumor tissue. Tissue specimens are then categorized into one of five groups based on the distribution of these patterns in the tumor. These groups are called International Society of Urological Pathology (ISUP) grade groups (Epstein et al., 2016; van Leenders et al., 2020).

4.1. Model development and internal testing

For model development and internal testing, we used the Prostate cANcer graDe Assessment (PANDA) dataset, introduced in Bulten et al. (2022) as part of a computational pathology challenge hosted on the Kaggle platform. It is the largest publicly available dataset of HE-stained prostate WSIs to date, with 11,554 biopsies curated from two different sites. During the competition phase, 10,616 biopsies were available for model development (the development set), and 393 biopsies were used as a public test set, facilitating model evaluation and fostering model comparisons among participants. When the development phase ended, teams could pick two models to be run on an additional 545 biopsies (the private test set), setting the final competition ranking.

Cases in the development and testing sets originate from two centers: the Radboud University Medical Center in Nijmegen (Netherlands) and the Karolinska Institutet in Stockholm (Sweden). The reference standards for the Radboud development set were based on the pathology reports from routine clinical practice. Karolinska development set was labeled by one uropathologist (L.E.) following routine clinical workflow. For the test sets, biopsies from Radboud were scored by a consensus of three expert uropathologists, while the Karolinska subset was collectively assessed by a panel of four highly experienced uropathologists.

Slides from Radboud University Medical Center were scanned using a 3DHistech Pannoramic Flash II 250 scanner at a pixel resolution 0.24 mpp, while slides from Karolinska Institutet were scanned at 20x magnification using two scanners: an Hamamatsu C9600-12 and an Aperio ScanScope AT2. The pixel resolution was 0.45 mpp (Hamamatsu) or 0.50 mpp (Aperio). In order to standardize resolution across centers and minimize the aggregate size of the different sets, the Radboud images were downsampled and provided at a pixel spacing of 0.48 mpp.

4.2. Generalization to independent test cohorts

Variations in tissue processing, slide preparation, and digitization can lead to significant differences in the appearance of WSIs. Consequently, it is crucial to ensure that the design choices we make generalize to real-world clinical data from scanners and staining protocols that are not encountered during training. We use the following two datasets as independent test cohorts to evaluate the generalization performance of our best model. The key characteristics of each test cohorts are summarized in Appendix F.

Karolinska university hospital dataset. This dataset was sourced from the Karolinska University Hospital. It was compiled in 2018 by L.E. and includes biopsy cores specifically selected to cover a range of ISUP grade groups (Bulten et al. (2022), Supplementary Methods). This careful selection resulted in 330 slides derived from 73 men. Cases were reviewed by a single uropathologist (L.E.) through a microscope. Slides were then scanned using a Hamamatsu C13220-01, ensuring high-resolution imaging at 0.46 mpp. Despite being private, this dataset was used as an external validation set to evaluate the best performing methods from the PANDA challenge, allowing for direct comparison.

Crowd-sourced dataset. Faryna et al. (2024) called through social media platforms to gather a diverse and representative collection of prostate biopsy slides from routine clinical practice. The resultant dataset comprises 113 cases collected from 7 centers, digitized using 5 different scanners. A single representative biopsy per case was selected by a pathology resident (LT) under the supervision of a subspecialized uropathologist (JvI). These biopsies were cropped and saved at a resolution of 0.50 mpp. A reader study involving 10 pathologists established the reference standard. For each case, the final label was assigned through a two-step process: first, the sample was categorized as benign or cancerous based on the 10 grades. If deemed cancerous, a majority vote among the cancer grades determined the final label. In case of ties, the highest Gleason grade group was assigned. This dataset is publicly available on the Grand Challenge platform (https://gleason-grading-in-the-wild.grand-challenge.org).

5. Experiments

5.1. Data preprocessing

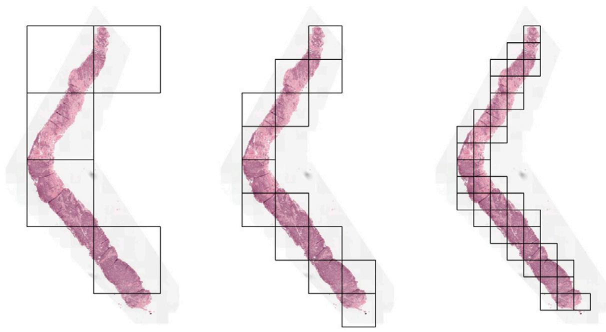
We adapted Lu et al. (2021) preprocessing pipeline to automatically segment tissue in each slide, from which we extract non-overlapping regions at the resolution closest to 0.50 mpp. To assess the influence of context size (Section 5.5), we not only extract 4096 × 4096 regions as in Chen et al. (2022), but also 1024 × 1024 and 2048 × 2048 regions. To account for tissue segmentation imprecision, we only retain regions containing over 10% tissue and save them as JPEG images, using Pillow's default compression quality setting of 75. Fig. 2 shows example results of region extraction for different region sizes. We additionally show the distribution of the number of regions extracted per slide, for different region sizes, in Appendix I. The average number of regions extracted per slide in PANDA is summarized in Table E.7.

5.2. Dataset splitting

5.2.1. Grouping similar looking slides

During the development phase of the PANDA challenge, many participants voiced concerns about having high internal tuning scores but low public leaderboard scores (Bulten et al. (2022), Supplementary Methods). PANDA challenge organizers acknowledged that a significant subset of the development data consists of serial sections coming from the same tissue blocks. This results in similar-looking slides, which have the potential to skew internal tuning.

Indeed, consider a scenario where two closely resembling slices from the same block, denoted as S_{train} and S_{tune}, become distributed between the training and tuning sets. Under such circumstances, the model gains some form of familiarity with slice S_{tune} during the training phase, due to its exposure to slice S_{train}. As a result, internal tuning no longer offers an unbiased evaluation of the model fit on the training dataset. To address this bias, splitting the data at the block level would ensure that slides originating from the same block are grouped within the same partition, be it the training or the tuning set. Unfortunately, the PANDA dataset lacks a direct slide-to-block mapping. Nevertheless, we design an algorithm to create a hypothetical mapping that can later be used for splitting the data.



(a) (b) (c)

Fig. 2. Visualization of region extraction at 0.50 mpp for varying region sizes: **(a)** (4096, 4096) regions, **(b)** (2048, 2048) regions and **(c)** (1024, 1024) regions.

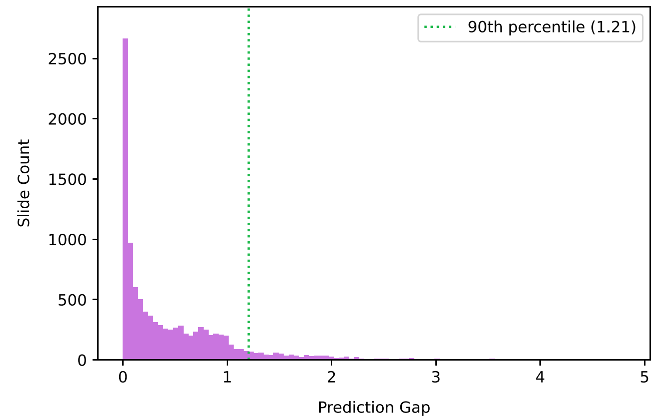


Fig. 3. Distribution of the prediction gap for PANDA development set.

Table 1
Scanner details, PANDA development set.

Center	Scanner vendor	# slides
Radboud	3DHistech	5160
Karolinska	Leica	2193
Karolinska	Hamamatsu	3263

We start by computing image similarity for each pair of slides using image hashing, a technique that generates a hash uniquely identifying an image based on its content. We leverage the open-source Python library `imagehash` for the implementation. After computing hash similarity for each pair of slides, we eliminate pairs with a similarity score below a preset threshold, τ_{sim}. Since a patient cannot have slides scanned with two different scanners, we remove some false positives by excluding matches across different scanners. We build an undirected graph with the remaining pairs where nodes are slides, and edges are hash similarities. Scanning for connected components allows identifying potential blocks. In the end, using $\tau_{\text{sim}} = 0.94$ enables us to identify a sufficient number of similar-looking slides while maintaining a reasonable cap on the maximum number of slides grouped into a single block. We obtain 297 connected components, covering 642 unique slides. This roughly amounts to 6% of the development set. We show example blocks created with our algorithm in Appendix H.

5.2.2. 5-Fold cross validation

The large size of the PANDA dataset allows data partitioning through stratification on a few variables, effectively curbing potential sources of bias. We split the development set into three distinct buckets based on the originating center and the scanner vendor (Table 1). A block-to-grade mapping is necessary to split each data bucket at the block level by stratifying on ISUP grade. We build such a mapping by assigning each artificial block its prevailing ISUP grade — referred to as `isup_grade_maj`. We break ties by randomly assigning one of the tied prevailing grades. Each bucket is independently split into 5 cross validation folds, stratifying on `isup_grade_maj` (Fig. G.2(a)). For each bucket, 20% of the training set is sub-sampled for model tuning, again stratifying on `isup_grade_maj`. This was necessary for label denoising (Section 5.2.3). The resulting folds from each bucket are eventually merged to form the final 5-fold cross validation sets (Fig. G.2(b)).

5.2.3. Label denoising

As detailed in Bulten et al. (2022) supplementary methods, label reporting tasks are carried out manually, leading to a certain degree of label noise in PANDA development data. After splitting the development set into five folds, we train a distinct model on each fold and run inference on the corresponding hold-out test set. Preliminary experiments revealed that training a Local H-ViT on (1024, 1024) regions achieved quadratic weighted kappa scores that closely aligned with the highest ranked solutions in the public leaderboard (Table B.4). This configuration was deemed satisfactory for cleaning up label noise and was subsequently used across all folds.

Using the cross-entropy loss, each slide gets assigned a soft prediction, obtained as the class-weighted sum of probabilities. Soft predictions better convey the model uncertainty, offering a refined prediction spectrum. We compute the prediction gap between soft predictions and groundtruth labels (Fig. 3) and exclude the 10% slides most likely to be noisy by dropping samples whose gap is larger than $\tau_{\text{gap}} = 1.21$.

After removing the top 10% of samples with the highest discrepancies between model predictions and actual labels from the development set, we restructure our dataset configuration. For each fold, we combine the training and tuning partitions to form a consolidated training set, thereby maximizing the amount of data used for learning. The testing partition of each fold, previously used to get model predictions, now serves as the new tuning set for hyperparameter optimization (Fig. G.2(b)). This updated configuration forms the *cleaned* 5-fold cross-validation split. Unless stated otherwise, all subsequent experiments are carried out on this configuration. Final model evaluation is conducted on the official PANDA test sets, allowing for comparison with challenge participants. Table G.9 provides a detailed breakdown of the number of slides in each partition for the cleaned split.

5.3. Self-supervised pretraining

We begin by evaluating the effectiveness of the pretrained weights provided by Chen et al. (2022), who pretrained both patch-level and region-level Transformers using 10,678 HE-stained slides covering 33 cancer types from The Cancer Genome Atlas (TCGA). As shown in Table 2, these weights demonstrate limited performance on the PANDA public test set, with quadratic weighted kappa scores ranging from 0.42 to 0.62. This limited performance can be attributed to the relatively small proportion of prostate samples in the pretraining data—only 4% of the TCGA slides are prostate resections. This calls for more targeted pretraining to better capture the distinct pathological features of prostate cancer in biopsy samples.

Therefore, we investigate the impact of pretraining both Transformers exclusively on prostate biopsies from the PANDA dataset. To avoid data leakage from the tuning set, we pretrain a separate model for each fold in our cross-validation setup. Pretraining is performed on the training set of each fold, before label denoising, and the same pretraining strategy is applied across all folds. We pretrain the patch-level Transformer for 50 epochs, using early stopping based on the downstream evaluation dataset. Then, we pretrain the region-level

Table 2
Classification performance of Global H-ViT for different pretraining configurations, obtained with cross-entropy loss. We report the mean and standard deviation of the quadratic weighted kappa across the 5 cross-validation folds, along with the kappa score achieved by ensembling predictions from each fold.

Region size	Pretraining dataset	Tune	Test (public)	Test (public, *ens.*)
4096	TCGA	0.815 ± 0.010	0.422 ± 0.077	0.4264
4096	PANDA	**0.853 ± 0.019**	**0.721 ± 0.066**	**0.7904**
2048	TCGA	0.822 ± 0.005	0.416 ± 0.079	0.4353
2048	PANDA	**0.871 ± 0.018**	**0.798 ± 0.035**	**0.8234**
1024	TCGA	0.831 ± 0.012	0.599 ± 0.066	0.6243
1024	PANDA	**0.924 ± 0.010**	**0.844 ± 0.018**	**0.8742**

Transformer for 125 epochs. Appendix J provides more details on the computational requirements of pretraining both Transformers.

Pretraining of the patch-level Transformer on TCGA by Chen et al. (2022) represents one of the earliest efforts to develop a generic vision encoder for pathology. Although the term *foundation model* had not yet been coined at the time of the original publication, this could be considered a small-scale foundation model. To further investigate the benefits of prostate-specific pretraining over more generic multi-organ pretraining, we conduct an additional experiment using UNI (Chen et al., 2024), a larger-scale foundation model. UNI is a ViT-Large pretrained on over 100,000 HE-stained slides from a wide range of tissue types and diseases. It has recently been recognized as one of the leading publicly available foundation models in histopathology (Campanella et al., 2024).

5.4. Model training

5.4.1. Baseline

We replicate the best configuration presented in Chen et al. (2022) as a baseline: Global H-ViT with TCGA-pretrained weights for the patch-level and region-level Transformers. Slides in the PANDA datase thus get encoded into a (M_i^{4096}, 192) *global* feature vector, where M_i^{4096} stands for the number of (4096, 4096) regions extracted in the ith slide. These vectors are then fed as input to the model, which is trained to generate a slide-level feature that is projected to class logits for loss computation. This results in roughly 0.6 million trainable parameters.

5.4.2. H-vit variants

We further experiment with Global H-ViT and Local H-ViT by varying the pretrained weights (TCGA-derived or PANDA-derived), the context size (Section 5.5) and the loss function (Section 5.6).

For Global H-ViT, regions of size (2048, 2048) (resp. (1024, 1024)) get encoded into 192-dimensional embeddings, resulting in (M_i^{2048}, 192) (resp. (M_i^{1024}, 192)) *global* feature vectors. The slide-level Transformer is then trained to pool these vectors into slide-level representations, which are subsequently mapped to the target labels via a linear layer. This also results in approximately 0.6 million trainable parameters, making it well-suited for situations where the number of labeled samples for weak supervision is limited, thus reducing the risk of overfitting.

For Local H-ViT, we leverage the pretrained patch-level Transformer to embed each non-overlapping (256, 256) patch in (4096, 4096) regions into 384-dim embeddings, such that each slide gets encoded into a (M_i^{4096}, 256, 384) *local* feature vector. Similarly, when using regions of size (2048, 2048) and (1024, 1024), slides are encoded into (M_i^{2048}, 64, 384) and (M_i^{1024}, 16, 384) embeddings, respectively. In this model variant, both the region-level and slide-level Transformers are trained to pool *local* feature vectors into 192-dim slide-level representations, which are then mapped to the target labels via a linear layer. This results in approximately 3.5 million trainable parameters. While Local H-ViT is more computationally intensive to train than Global H-ViT it has a higher capacity to learn complex mappings between input features and labels, potentially leading to improved classification performance. Therefore, we recommend Local H-ViT when sufficient training data is available. Appendix J provides more details on the computational characteristics of both model variants.

5.4.3. Experimental setup

All models are trained for 200 epochs using the Adam optimizer. We stop training early if the tuning loss does not improve for 20 consecutive epochs, with at least 50 epochs completed. We use a batch size of 1 with 32 gradient accumulation steps, a base learning rate of $2 \cdot 10^{-4}$. that gets decayed by a factor 2 every 20 epochs, and a weight decay of 10^{-5}.

5.5. Using smaller context size

Prostate cancer biopsies are usually smaller than surgical resections. This property translates into a smaller number of (4096, 4096) regions per slide. On average, slides from the PANDA dataset contain $M_{\text{PANDA}} \sim 6$ such regions. In contrast, Chen et al. (2022) report $M_{\text{TCGA}} \sim 38$ (4096, 4096) across the 10,678 slides from TCGA, which include 421 prostate slides, all of which are resections. Yet, the essence of the Transformer architecture lies in its ability to make sense of sequential input through multiple layers of self-attention. Providing too short sequences may not fully leverage Transformers learning capacities. Therefore, we experiment with smaller region sizes: (2048, 2048) and (1024, 1024) regions.

Despite these smaller regions, we intentionally keep the patch size (256) and token size (16) fixed, as these values were found to be optimal for the ViT architecture through extensive experimentation in Dosovitskiy et al. (2021). Changing these values would also alter the scale of features learned by the first two Transformers, which would interfere with our primary objective: determining the most effective context size.

We pretrain the region-level Transformer for each region size independently, as it yields better results than using the pretrained weights obtained with (4096,4096) regions and interpolating the position embedding L. For this, we adapt DINO data augmentation by taking (3,3) and (7,7) local–global crops for (M_i^{2048}, 8, 8, 384) inputs, and (2,2) and (4,4) local–global crops for (M_i^{1024}, 4, 4, 384) inputs, matching the scale of (96,96), (224, 224) crops for (256, 256) inputs.

As shown in Fig. 2, using smaller regions produces a finer-grained tiling of the slides. With fewer background pixels captured in each region, the signal-to-noise ratio increases. We hypothesize that this property facilitates model training.

When no regions can be extracted from a PANDA test set slide, we assign it a random prediction to ensure fair comparison with the challenge leaderboard.

5.6. Leveraging label order

When dealing with multiple classes, a common approach consists in treating each class as independent. In that case, classes are one-hot encoded. People typically use softmax activation, in conjunction with the categorical cross entropy (CE) loss. By treating classes as a set of unordered values, the CE loss equally penalizes wrong predictions. However, ISUP grades are ordered: the more aggressive the cancer, the higher the grade. By leveraging this order, we can provide the model with more information during training, potentially improving its predictive abilities.

Cheng (2007) introduced an effective approach to adapt a neural network to learn ordinal categories, which simply consists of changing how classes are encoded. When coupled with sigmoid activation and binary cross entropy (BCE) loss, this setup ensures the loss gets lower when the model gets closer to the correct class. To infer predictions, the method scans the model output from left to right and stops when it encounters a logit smaller than a predefined threshold τ_{class}. The index k of the last logit which is bigger than τ_{class} is the predicted class for that sample. We use $\tau_{\text{class}} = 0.5$ and refer to this setting as *ordinal CE*.

Unfortunately, nothing compels the model to output decreasing probabilities that match the regular pattern of the newly encoded labels, which is undesirable for making predictions. The following situation may arise where the output vector is [0.67, 0.86, 0.22, 0.91, 0.09]. Using $\tau_{\text{class}} = 0.5$, the inferred prediction would be class 2, when in fact the model is unsure between class 2 and class 4.

To resolve these inconsistencies, we experiment with two methods that offer theoretical guarantees for rank-consistency. The first method, COnsistent RAnk Logits (CORAL), is an architecture-agnostic framework that achieves rank-monotonicity by imposing a weight sharing constraint on the final linear layer, while allowing different biases (Cao et al., 2020). Although this approach ensures rank-consistency, the weight sharing limits the model's expressiveness. To address this limitation, we employ Conditional Ordinal Regression for Neural Networks (CORN), a more recent rank-consistent ordinal regression framework that does not impose any architectural constraints (Shi et al., 2021). CORN uses a new training procedure with conditional training subsets that ensures rank consistency through applying the chain rule of probability. Removing the weight-sharing constraint should improve the predictive performance of the model.

Eventually, we also formulate the classification problem as a regression task and use the Mean Squared Error (MSE) loss, which does leverage the ordinal nature of the labels. We conduct a series of experiment to determine which of these 5 setups – CE loss with one-hot labels, ordinal CE, CORAL, CORN, and MSE – works best in the context of ISUP grade prediction.

5.7. Evaluation metrics

The different models developed in this work are evaluated using Cohen's quadratic weighted kappa. This metric measures the agreement between two outcomes, typically ranging from 0 (random agreement) to 1 (complete agreement). Since all experiments were conducted via 5-fold cross-validation, we report the average quadratic weighted kappa scores for tuning and testing across the 5 folds. We additionally provide a single test quadratic weighted kappa per model configuration by ensembling predictions from each fold through majority voting (referred to as *ens.*). We explain our approach for handling ties during model ensembling in Appendix K.

We provide confusion matrices of our best ensemble model in Appendix C. To further assess model performance, we report two additional metrics for our best ensembled model in Appendix D: the overall classification accuracy, as well as the classification accuracy for distinguishing low-risk cases (ISUP ≤ 1) from higher-risk cases, which is clinically relevant for decision-making (Epstein et al., 2016).

6. Results

6.1. Self-supervised pretraining

Pretraining the patch-level Transformer for 50 epochs took 3 days on 4 GeForce RTX 3080 Ti. Fig. 4 shows the area under the curve (AUC) for teacher and student networks on the downstream patch-level classification dataset over pretraining epochs. Results are averaged across the 5 cross validation folds. Early stopping was not triggered for any of the 5 folds. Pretraining the region-level Transformer on

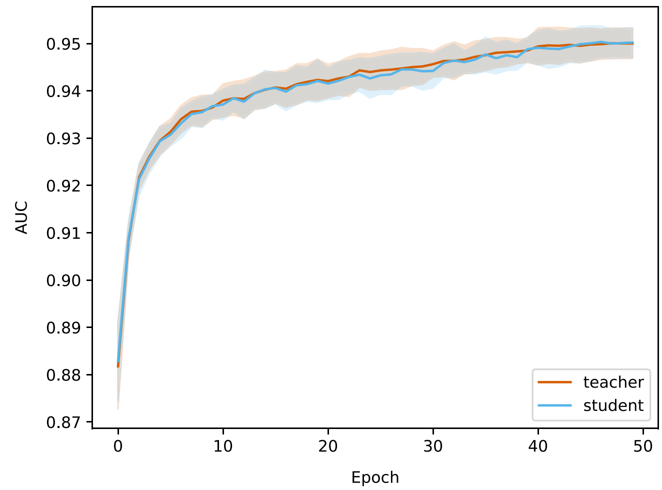


Fig. 4. Classification performance for teacher and student networks on the binary classification of prostate patches, used as downstream evaluation during pretraining. Lines represent the mean AUC across the 5 cross validation folds, with shaded areas indicating standard deviation.

(4096, 4096) regions on 1 GeForce RTX 3080 Ti. More details about computational characteristics can be found in Appendix J.

Classification results with CE loss are summarized in Table 2 (Global H-ViT) and Table 3 (Local H-ViT). Additional results for other loss functions can be found in Appendix A (Global H-ViT) and Appendix B (Local H-ViT). Overall, across all region sizes, models pretrained on the PANDA dataset consistently achieve higher macro-averaged performance than those pretrained on TCGA. Performance gains are more significant for Global H-ViT (+68% on average) than for Local H-ViT (+14% on average). This roots back to a common limitation of patch-based MIL: the disconnection between feature extraction and feature aggregation. There is no guarantee the features extracted during pretraining are relevant for the downstream classification task. Allowing gradients to flow through the region-level Transformer partly overcomes this limitation: Local H-ViT has more amplitude than Global H-ViT to refine the TCGA-pretrained features so that they better fit the classification task at hand.

6.2. Using smaller context size

Our results show that context size and model variant are closely entangled. Global H-ViT starts from features that are already aggregated at the region level. In that case, the context size directly affects the amount of information summarized: tiling the slide with regions that are twice smaller can result in up to four times more regions. Since each region is embedded into a 192-dimensional feature vector, regardless of its size, the amount of information captured in the features passed to Global H-ViT can be up to four times bigger. This could explain why using (1024, 1024) regions achieves the highest macro-averaged scores for Global H-ViT (Table 2, Appendix A). Performance gains are less obvious for Local H-ViT, where (2048, 2048) and (1024, 1024) regions equally work slightly better than (4096, 4096) regions. Since Local H-ViT starts from features prior to region-level aggregation, the amount of information is determined by both the region size and the patch size. As the patch size is kept fixed, the amount of information passed to the slide-level Transformer is roughly the same across region sizes, resulting in less pronounced differences. Nevertheless, (2048, 2048) regions work best B.

Overall, we observe improved performances when working with smaller context sizes. Unlike other prediction tasks that may benefit from broader context, ISUP grading does not require integrating a lot of context as Gleason patterns typically fit within the boundary of (1024,

Table 3
Classification performance of Local H-ViT for different pretraining configurations, obtained with cross-entropy loss. We report the mean and standard deviation of the quadratic weighted kappa across the 5 cross-validation folds, along with the kappa score achieved by ensembling predictions from each fold.

Region size	Pretraining dataset	Tune	Test (public)	Test (public, *ens.*)
4096	TCGA	0.803 ± 0.017	0.761 ± 0.018	0.7732
4096	PANDA	**0.936** ± **0.008**	**0.882** ± **0.016**	**0.8809**
2048	TCGA	0.814 ± 0.014	0.777 ± 0.023	0.8044
2048	PANDA	**0.942** ± **0.004**	**0.882** ± **0.011**	**0.8976**
1024	TCGA	0.814 ± 0.011	0.774 ± 0.012	0.7926
1024	PANDA	**0.943** ± **0.005**	**0.878** ± **0.024**	**0.8911**

Table 4
Classification performance of Global H-ViT for different loss functions, obtained with (1024, 1024) regions at 0.50 mpp. We report the mean and standard deviation of the quadratic weighted kappa across the 5 cross-validation folds, along with the kappa score achieved by ensembling predictions from each fold.

Loss	Tune	Test (public)	Test (public, *ens.*)
CE	0.924 ± 0.010	0.844 ± 0.018	0.8742
ordinal CE	0.929 ± 0.006	0.857 ± 0.013	0.8642
CORAL	0.901 ± 0.007	0.827 ± 0.014	0.8534
CORN	0.924 ± 0.006	0.857 ± 0.017	0.8683
MSE	**0.930** ± **0.004**	**0.864** ± **0.011**	**0.8784**

Table 5
Classification performance of Local H-ViT for different loss functions, obtained with (2048, 2048) regions at 0.50 mpp. We report the mean and standard deviation of the quadratic weighted kappa across the 5 cross-validation folds, along with the kappa score achieved by ensembling predictions from each fold.

Loss	Tune	Test (public)	Test (public, *ens.*)
CE	0.942 ± 0.004	0.882 ± 0.011	0.8976
ordinal CE	0.948 ± 0.004	0.894 ± 0.012	0.9053
CORAL	0.896 ± 0.006	0.859 ± 0.008	0.8782
CORN	0.941 ± 0.007	0.901 ± 0.011	0.9077
MSE	**0.950** ± **0.002**	**0.904** ± **0.011**	**0.9149**

1024) regions at 0.50 mpp. This aligns with our experimental findings. Moreover, working with smaller regions results in finer grained slide tiling: the proportion of non-informative background pixels in each region is reduced, as illustrated in Fig. 2. This results in a higher signal-to-noise ratio, which eases model training.

6.3. Leveraging label order

The results of the experiments with different loss functions are summarized in Table 4 (Global H-ViT) and Table 5 (Local H-ViT). CORAL consistently underperforms, confirming that the weight-sharing constraint it imposes in the last linear layer restricts the model's expressiveness. CORN, introduced to overcome this limitation, closes this performance gap. Notably, standard cross entropy loss consistently yields inferior results compared to ordinal CE, CORN, or MSE, indicating that leveraging the ordinal nature of ISUP labels guides the learning process towards better local optima. For both model variants, treating ISUP grade prediction as a regression task using MSE loss produces the highest macro-averaged quadratic weighted kappa scores.

6.4. H-ViT variants

Results on PANDA public test set show that Local H-ViT consistently outperforms Global H-ViT across all configurations (Tables 2, 3, 4, 5, Appendix A, Appendix B). This finding is particularly noteworthy as it deviates from the conclusions presented in Chen et al. (2022). Yet, it aligns more closely with theoretical expectations: although Local H-ViT and Global H-ViT share the same architecture, their gradient flows differ significantly. In Global H-ViT, gradients are restricted to the slide-level Transformer. In contrast, Local H-ViT lets gradients flow through both the slide-level and region-level Transformers, allowing it to adjust a greater number of parameters and better fit the data. To further confirm this theoretical advantage, we replicate the breast cancer subtyping experiment from Chen et al. (2022). Again, our results show that Local H-ViT outperforms Global H-ViT (Table 6).

Table 6
TCGA BRCA subtyping results. Both models are trained with (4096,4096) regions on the splits from Chen et al. (2022). We report the mean and standard deviation of the AUC across the 10 cross-validation folds.

Model	Chen et al. (2022)	Ours
Global H-ViT	**0.874** ± **0.060**	0.887 ± 0.064
Local H-ViT	0.827 ± 0.069	**0.894** ± **0.049**

These results suggest that the best overall configuration is Local H-ViT, using features from our PANDA-pretrained encoder and trained with MSE loss on (2048, 2048) regions. We fix this configuration and conduct an additional experiment, replacing our PANDA-pretrained encoder with UNI. Results, summarized in Table 7, are comparable, suggesting UNI can serve as an effective alternative in this setting. We then proceed to generalization experiments with these two model configurations to determine which performs best when evaluated across diverse clinical settings.

6.5. PANDA test sets

To ensure a fair comparison with challenge participants, we limit our testing on PANDA private test set to a single run. We additionally run our model on PANDA combined test set, obtained by combining public and private test samples. Table 8 shows quadratic weighted kappa scores of our ensembled models against that of the top-performing teams invited to join the PANDA consortium. Best submissions demonstrated remarkable accuracy, achieving kappa scores ranging from 0.89 to 0.92 on the public test set. Similarly, on the private test set, the leading entries scored between 0.92 and 0.94. In our comparative analysis, a two-sided permutation test on the combined test set between our best ensemble model and the best-performing team

Table 7
Classification performance of Local H-ViT for different feature encoders, obtained with MSE loss and (2048, 2048) regions at 0.50 mpp. We report the mean and standard deviation of the quadratic weighted kappa across the 5 cross-validation folds, along with the kappa score achieved by ensembling predictions from each fold.

Patch-level Transformer	Tune	Test (public)	Test (public, *ens.*)
UNI	0.947 ± 0.005	0.901 ± 0.011	0.9039
PANDA-pretrained ViT-S	**0.950 ± 0.002**	**0.904 ± 0.011**	**0.9149**

(`Save The Prostate`) shows the difference in performance is not statistically significant (p = 0.1235). This demonstrates our best model achieves performance on par with the state-of-the-art on PANDA. Using UNI as the patch-level feature encoder performs on par with our best model on the PANDA sets, further indicating that UNI is a viable feature extractor for this dataset. The confusion matrix of our best ensemble model is shown in Fig. C.1(a). Additional accuracy metrics can be found in Appendix D.

Despite these encouraging results, the PANDA test sets are more akin to in-domain testing, given the slides used for model development originate from the same centers. This raises concerns about overly optimistic estimates of the model performance in routine clinical settings. To address this, we evaluate our model on two independent test cohorts.

6.6. Generalization to independent test cohorts

Quadratic weighted kappa scores on the Karolinska University Hospital dataset are summarized in the final column of Table 8. Our model ranks second among the 15 evaluated models. A two-sided permutation test shows that the difference in quadratic weighted kappa between our model and the best model (`NS_Pathology`) is not statistically significant (p = 0.9333). Fig. C.1(b) shows the confusion matrix of our best ensemble model on this dataset.

The appearance of slides in the Karolinska University Hospital slides is close to that of the PANDA dataset originating from the Karolinska Institutet. This could explain why most models generalize well on this dataset and calls for additional evaluation in a more diverse clinical setting. Because of its crowd-sourced nature, the prostate biopsy dataset introduced in Faryna et al. (2024) encompasses diverse clinical settings (tissue preparation protocol, staining, scanning device). It even includes slides stained with hematoxylin, eosin and saffron, a staining common in France but absent in PANDA development set. The diversity of this dataset allows for a more rigorous assessment of model performance in a real-world setting.

As shown in Table 9, our model performs on par or better than all other models across all reported metrics on the crowdsourced dataset. While most PANDA challenge participants models show a significant drop in performance, our best model maintains strong generalization. A two-sided permutation test between our model and the second-best performing team (`PND`) shows that the difference in quadratic weighted kappa is not statistically significant (p = 0.7313). Nonetheless, this result establishes a new benchmark in prostate cancer grading from biopsies, effectively narrowing the generalization gap. Fig. C.1(c) shows the confusion matrix of our best ensemble model on this dataset.

Using UNI as patch-level Transformer in Local H-ViT results in a significant performance drop on these two datasets. While it performs well on the PANDA dataset, the UNI-based model struggles to generalize in more diverse clinical settings, where a general-purpose foundation model is expected to excel. This indicates that the prostate-specific features learned by our PANDA-pretrained ViT-S generalize more effectively to unseen centers. These findings align with recent research indicating that pathology foundation models tend to learn center-specific representations rather than universal ones (de Jong et al., 2025), highlighting the value of prostate-specific pretraining in capturing the nuances of prostate pathology.

6.7. Interpretability

6.7.1. Individual attention heatmaps

Attention heatmaps offer a streamlined form of model interpretability by revealing the specific image features that the model has learned to associate with particular classes. A natural starting point for deriving a form of model explainability is to visualize the stitched attention heatmaps for each of the three Transformers individually (Fig. 5). Depending on the model configuration, these heatmaps may reveal either generic features learned during self-supervised pretraining, or task-specific features learned through later stage training.

However, individual heatmaps offer an incomplete view of our model's decision-making process. For a more comprehensive form of model interpretability, it becomes necessary to develop methods that effectively combine attention scores across the entire hierarchical architecture.

6.7.2. Factorized attention heatmaps

When combining attention scores across hierarchical Transformers, we introduce the parameter γ to control how attention scores from self-supervised pretraining are merged with those from supervised training (Eq. (2)). By tuning γ, we can adjust the model's focus between task-agnostic, self-supervised features and task-specific, weakly-supervised features, as well as between fine-grained, cell-level details and coarse-grained, region-level patterns. As a result, the factorized heatmaps generated through this approach provide a more comprehensive view of model attention compared to those derived from individual Transformers.

In Global H-ViT, both patch-level and region-level Transformers are frozen (n=2). Setting $\gamma = 0$ in Eq. (2) nullifies the contribution of the (trained) slide-level Transformer, giving equal importance to the patch-level and region-level attention. This specific case corresponds to the hierarchical heatmap implementation presented in Chen et al. (2022):

$$a_{(x,y)} = \frac{1}{2}\left(a^0_{(x,y)} + a^1_{(x,y)}\right) \tag{3}$$

Yet, this approach limits the analysis to features acquired during pretraining. Given these may not necessarily align with ISUP grades, it greatly narrows the insights that can be derived from the heatmaps. By choosing $\gamma > 0$, we also incorporate attention scores from the slide-level Transformer, which directly relate to ISUP grades, enriching the analysis. When $\gamma = 0.5$, both trained and frozen Transformers contribute equally to the factorized attention score. When $\gamma = 1$ the frozen layers' attention is silenced out, such that the attention score solely comes from the slide-level Transformer.

In Local H-ViT, only the patch-level Transformer is frozen (n=1). When $\gamma = 0$, the attention score is exclusively determined by the (frozen) patch-level Transformer. Instead, setting $\gamma = 1$ in Eq. (2) nullifies the contribution of the (frozen) patch-level Transformer, equally weighting the region-level and slide-level attention scores:

$$a_{(x,y)} = \frac{1}{2}\left(a^1_{(x,y)} + a^2_{(x,y)}\right) \tag{4}$$

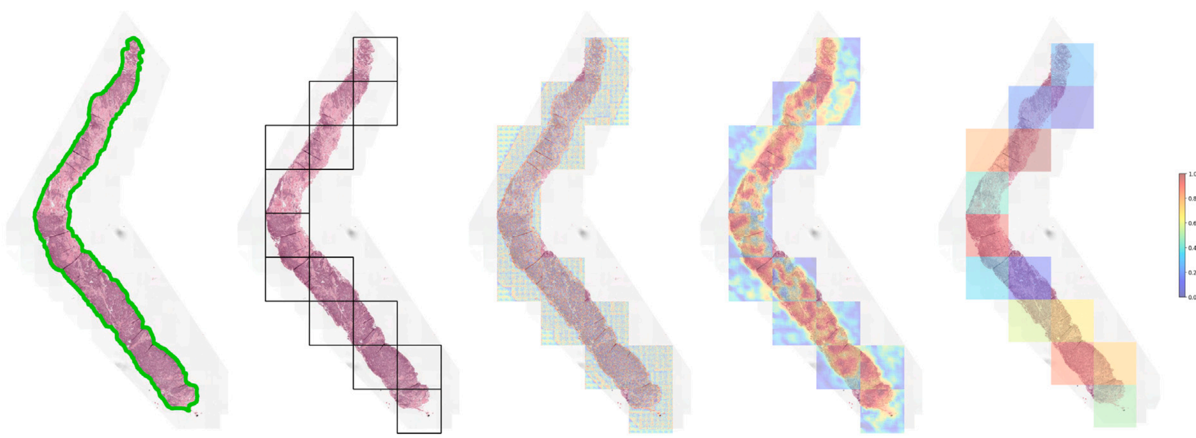
Finally, when $\gamma = 0.5$, we fallback onto the case where the contributions from frozen and trained Transformers are equal.

There is no single optimal value for γ, but medical knowledge can guide the selection of meaningful values. For instance, the morphologic

Table 8
Classification performance of our best ensemble Local H-ViT models against that of PANDA consortium teams on PANDA public and private test sets, as well as Karolinska University Hospital dataset, used as external validation data after the challenge ended. All values are given as quadratic weighted kappa.

Model	Test (public)	Test (private)	Test (public+private)	Karolinska University Hospital dataset
Save The Prostate	0.9209	**0.9377**	**0.9280**	0.881
NS Pathology	0.9180	0.9340	**0.9272**	**0.899**
PND	0.9108	**0.9408**	0.9252	0.890
iafoss	0.9179	0.9301	0.9250	0.824
Aksell	**0.9212**	0.9274	0.9247	0.879
vanda	**0.9219**	0.9303	0.9215	0.880
ChienYiChi	0.9086	0.9324	0.9214	0.886
BarelyBears	0.9118	0.9326	0.9204	0.890
rʿahmʾa.ai	0.9096	0.9262	0.9182	0.865
ctrasd123	0.8948	0.9324	0.9165	0.879
Kiminya	0.9007	0.9328	0.9164	0.881
Manuel Campos	0.8935	0.9296	0.9142	0.892
Dmitry A. Grechka	0.8861	0.9283	0.9105	0.740
KovaLOVE v2	0.8889	0.9277	0.9099	0.880
Local H-ViT[a] (*ens.*)	0.9149	0.9170	0.9161	**0.895**
Local H-ViT[b] (*ens.*)	0.9039	0.9151	0.9104	0.867

[a] Starting from features given by our PANDA-pretrained ViT-S.
[b] Starting from features given by UNI.



(a) tissue segmentation (b) slide tiling (c) patch-level (d) region-level (e) slide-level

Fig. 5. Visualization of stitched attention heatmaps for each Transformer. We show the result of tissue segmentation **(a)** where tissue is delineated in green, and the result of slide tiling into non-overlapping (2048, 2048) regions as 0.50 mpp, keeping only regions with 10% tissue or more. For each of the three Transformers, we overlay the corresponding attention scores assigned to each element of the input sequence: (16, 16) tokens for the patch-level Transformer, (256, 256) patches for the region-level Transformer, and (2048, 2048) regions for the slide-level Transformer.

Table 9
Classification performance of our ensemble Local H-ViT models compared to five PANDA consortium teams on the crowdsourced dataset. We report quadratic weighted kappa (κ^2), overall accuracy (acc), and binary accuracy (bin-acc) for distinguishing between low-risk (ISUP $\leq$ 1) and higher-risk cases.

Model	κ^2	acc	bin-acc
PND	**0.862**	0.513	0.876
BarelyBears	0.845	0.531	**0.903**
NS Pathology	0.760	**0.611**	0.876
Kiminya	0.716	0.513	0.867
vanda	0.617	0.336	0.779
Local H-ViT[a] (*ens.*)	**0.877**	**0.602**	**0.903**
Local H-ViT[b] (*ens.*)	0.793	0.575	0.832

[a] Starting from features given by our PANDA-pretrained ViT-S.
[b] Starting from features given by UNI.

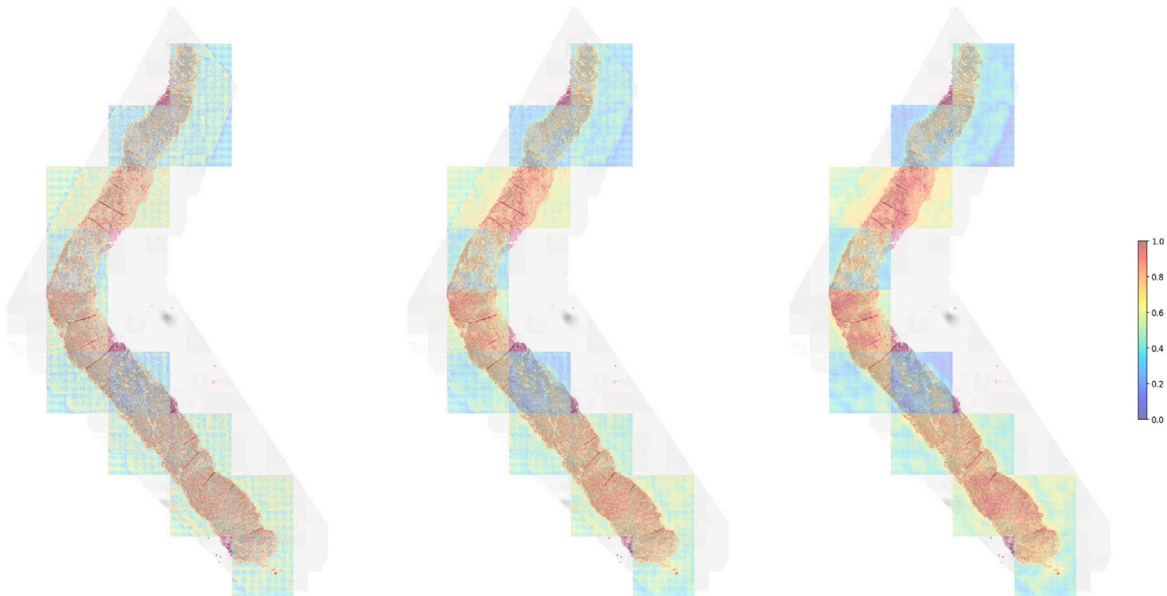
features essential for prostate cancer grading are found at the level of the glandular architecture of the tissue, at a much larger scale than cell-level details. Since the patch-level Transformer is restricted to capturing cell-level features, we recommend using $\gamma > 0.5$ to emphasize coarser, region-level features (Fig. 6).

7. Discussion

This work explored the use of Hierarchical Vision Transformers (H-ViTs) for predicting prostate cancer grades from biopsy whole-slide images. We focused on four critical aspects: self-supervised pretraining, model variant, context size, and the choice of loss function. By conducting independent testing, we aimed to reduce potential biases and identify the optimal model configuration.

Our experiments demonstrated that organ-specific pretraining consistently outperforms multi-organ pretraining. First, we compared the performance of TCGA-pretraining, as used by Chen et al. (2022), with pretraining exclusively on the PANDA dataset. As shown in our results, the PANDA-pretrained H-ViTs outperform the TCGA-pretrained ones across all experiments, highlighting the importance of domain-specific pretraining for prostate cancer pathology. The limited performance of the TCGA-pretrained H-ViT on the PANDA dataset can be attributed to the relatively small proportion of prostate samples in the TCGA pretraining data—only 4% of the TCGA slides are prostate resections.

To further investigate the impact of organ-specific pretraining, we experimented with UNI, a large-scale foundation model pretrained over 100,000 HE-stained slides from various tissue types and diseases. Despite achieving strong performance on the PANDA test sets,



(a) $\gamma = 0.3$ (b) $\gamma = 0.5$ (c) $\gamma = 0.7$

Fig. 6. Refined (2048,2048) factorized attention heatmaps of Local H-ViT for varying values of parameter γ. In the context of prostate cancer grading, the relevant signal is found within the tissue architecture, spanning intermediate to large scales. Since the patch-level Transformer primarily captures cell-level features, we recommend using $\gamma > 0.5$ to emphasize coarser, task-specific features, rather than finer, task-agnostic details.

the UNI-based H-ViT significantly underperforms on the more diverse crowdsourced dataset, where a general-purpose foundation model is expected to excel. This indicates that the prostate-specific features learned by our PANDA-pretrained ViT-S generalize more effectively across unfamiliar centers, while UNI's features seem more prone to overfitting center-specific signals. We hypothesize that UNI's center-specific bias leads to a mismatch between the features seen during training and those extracted during inference. Even large-scale, multi-organ pretraining may not capture the unique characteristics of prostate cancer pathology in biopsy samples, emphasizing the importance of domain-specific pretraining.

We compared two model variants, Global and Local H-ViT, and provide evidence-based guidelines for selecting the appropriate model. Specifically, Global H-ViT is more parameter-efficient and better suited for smaller datasets, where the benefits of self-supervised pretraining the region-level Transformer outweigh the cost of training both the region-level and slide-level Transformers. However, when sufficient labeled data is available, Local H-ViT's larger learning capacity results in better performance.

Our results further suggest that the choice of context size has a limited effect in prostate biopsy grading, as the critical signal is already contained within smaller regions. For Global H-ViT, using smaller regions increases the amount of summarized information, enabling more effective encoding of tissue architecture. In contrast, Local H-ViT, which processes features before region-level aggregation, shows minimal performance variation across context sizes.

Additionally, we demonstrated the superiority of treating prostate cancer grading as a regression task. Harnessing the ordinal nature of the labels through MSE loss facilitated the learning process. Finally, we introduced a novel technique for combining attention across hierarchical Transformers. This innovative approach allows for an intricate balance between the attention scores originating from self-supervised pretraining and those computed while training the model on a specific task.

Generalization experiments showed that top-performing models from the PANDA challenge often struggled to maintain performance when evaluated in more diverse clinical settings, highlighting the bias models may acquire when trained and tested in similar clinical environments. Notably, our best model matched state-of-the-art algorithms on in-domain data while consistently outperforming them on out-domain data. This result sets a new benchmark in prostate biopsy cancer grading, narrowing the generalization gap.

Despite these promising results, our work has limitations that offer avenues for future research. We only included augmentation during pretraining, and sticked to DINO's settings, which were designed for natural images. Incorporating augmentations tailored for pathology images could enhance self-supervised feature learning. Furthermore, we did not use any data augmentation during the weakly supervised training phase. Augmentation in the pixel space are computationally expensive, as each augmented region needs to be embedded independently. Applying transformations directly in the feature space – such as adding noise or interpolating between features (DeVries and Taylor, 2017) or using advanced methods like the data augmentation generative adversarial network (Zaffar et al., 2022) – could mitigate this issue.

Our model would best serve as a decision-support tool rather than a standalone diagnostic system, adding an extra layer of confidence for pathologists who already have established workflows. Attention heatmaps could be used to highlight regions of interest within biopsy slides and potentially streamline the review process. However, integrating such models into clinical practice requires dedicated efforts: heatmaps alone do not fully elucidate how critical regions relate to specific labels. Pathologists may be cautious about adopting algorithmic tools without transparent reasoning processes. Future research should focus on improving model explainability. For example, deriving label-specific attention scores would allow pathologists to query the model to see which parts of an image correspond to a given label, thereby offering deeper insight into the model's decision-making process.

8. Conclusion

In summary, our study demonstrates the transformative potential of Hierarchical Vision Transformers in predicting prostate cancer grades from biopsies. By leveraging the inherent hierarchical structure of

WSIs, H-ViTs efficiently capture context-aware representations, addressing several shortcomings of conventional patch-based methods. Our findings set new benchmarks in prostate cancer grading. Our model outperforms existing solutions when tested on a dataset that uniquely represents the full diversity of cases seen in clinical practice, effectively narrowing the generalization gap in prostate biopsy grading. This work provides new insights that deepen our understanding of the mechanisms underlying H-ViTs' effectiveness. Our results showcase the robustness and adaptability of this method in a new setting, paving the way for its broader adoption in computational pathology and potentially other medical imaging tasks.

CRediT authorship contribution statement

Clément Grisi: Writing – review & editing, Writing – original draft, Methodology. **Kimmo Kartasalo:** Writing – review & editing. **Martin Eklund:** Writing – review & editing. **Lars Egevad:** Writing – review & editing. **Jeroen van der Laak:** Writing – review & editing. **Geert Litjens:** Writing – review & editing.

Ethics approval and consent to participate

For all data used in this paper, an ethical committee was consulted and waived the need for patient consent.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Geert Litjens reports financial support was provided by European Research Council. Geert Litjens reports a relationship with Aiosyn BV that includes: equity or stocks. Geert Litjens reports a relationship with Canon Health Informatics that includes: consulting or advisory. Jeroen van der Laak reports a relationship with Philips that includes: board membership and funding grants. Jeroen van der Laak reports a relationship with ContextVision that includes: board membership and funding grants. Jeroen van der Laak reports a relationship with Sectra that includes: funding grants. Jeroen van der Laak reports a relationship with Aiosyn BV that includes: employment and equity or stocks. Kimmo Kartasalo reports a relationship with Clinsight AB that includes: employment and funding grants. Kimmo Kartasalo reports a relationship with David and Astrid Hagelen Foundation that includes: funding grants. Kimmo Kartasalo reports a relationship with Karolinska Institute Research Foundation that includes: funding grants. Kimmo Kartasalo reports a relationship with KAUTE Foundation that includes: funding grants. Kimmo Kartasalo reports a relationship with Orion Research Foundation that includes: funding grants. Kimmo Kartasalo reports a relationship with Oskar Huttunen Foundation that includes: funding grants. Lars Egevad reports a relationship with Clinsight AB that includes: employment and equity or stocks. Martin Eklund reports a relationship with Clinsight AB that includes: employment and equity or stocks. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Global H-ViT results

See Tables A.1 and A.2.

Appendix B. Local H-ViT results

See Tables B.3 and B.4.

Appendix C. Confusion matrices

See Fig. C.1.

Appendix D. Additional performance metrics

See Tables D.5 and D.6.

Appendix E. Average number of extracted regions

See Table E.7.

Appendix F. Key characteristics of test cohorts

See Table F.8.

Appendix G. 5-Fold cross validation splits

See Table G.9 and Fig. G.2.

Appendix H. Artificial blocks

See Fig. H.3.

Appendix I. Region extraction results

See Fig. I.4.

Table A.1
Global H-ViT results when pretrained on TCGA dataset, for different region sizes and loss functions. We report the mean and standard deviation of the quadratic weighted kappa across the 5 cross-validation folds, along with the kappa score achieved by ensembling predictions from each fold.

Region size	Loss	Tune	Test (public)	Test (public, *ens.*)
4096	CE	0.815 ± 0.010	0.422 ± 0.077	0.4264
	ordinal CE	**0.840 ± 0.008**	0.422 ± 0.050	0.4389
	CORAL	0.831 ± 0.013	0.405 ± 0.066	0.4058
	CORN	0.821 ± 0.014	**0.542 ± 0.055**	**0.5612**
	MSE	0.837 ± 0.007	0.416 ± 0.075	0.4143
2048	CE	0.822 ± 0.005	0.416 ± 0.079	0.4353
	ordinal CE	0.843 ± 0.010	0.432 ± 0.085	0.4117
	CORAL	0.839 ± 0.009	0.471 ± 0.102	0.4682
	CORN	0.837 ± 0.009	**0.518 ± 0.099**	**0.5227**
	MSE	**0.851 ± 0.010**	0.443 ± 0.123	0.4027
1024	CE	0.831 ± 0.012	0.599 ± 0.066	0.6243
	ordinal CE	0.848 ± 0.009	0.649 ± 0.063	0.6893
	CORAL	0.847 ± 0.009	0.621 ± 0.094	0.6676
	CORN	0.849 ± 0.018	**0.679 ± 0.051**	**0.7073**
	MSE	**0.858 ± 0.013**	0.662 ± 0.070	0.6794

Table A.2
Global H-ViT results when pretrained on PANDA dataset, for different region sizes and loss functions. We report the mean and standard deviation of the quadratic weighted kappa across the 5 cross-validation folds, along with the kappa score achieved by ensembling predictions from each fold.

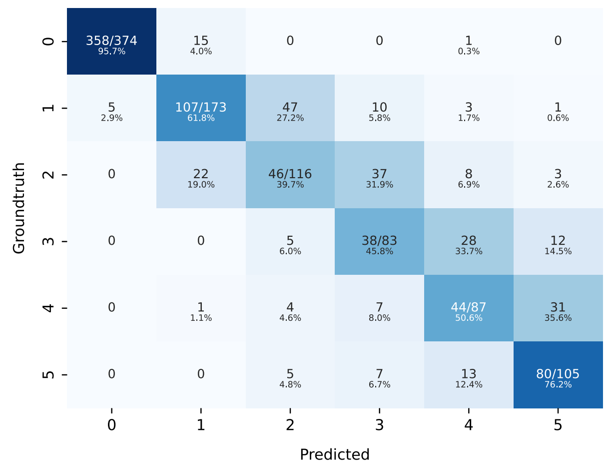
Region size	Loss	Tune	Test (public)	Test (public, *ens.*)
4096	CE	0.853 ± 0.019	0.721 ± 0.066	0.7904
	ordinal CE	**0.870 ± 0.022**	0.761 ± 0.048	0.7919
	CORAL	0.859 ± 0.014	0.723 ± 0.060	0.7703
	CORN	0.860 ± 0.015	**0.766 ± 0.052**	**0.7988**
	MSE	0.869 ± 0.020	0.747 ± 0.042	0.7939
2048	CE	0.871 ± 0.018	0.798 ± 0.035	0.8234
	ordinal CE	0.888 ± 0.022	0.783 ± 0.028	0.8118
	CORAL	0.870 ± 0.017	0.773 ± 0.021	0.8097
	CORN	0.880 ± 0.012	0.790 ± 0.033	0.8070
	MSE	**0.888 ± 0.019**	**0.799 ± 0.018**	**0.8243**
1024	CE	0.924 ± 0.010	0.844 ± 0.018	0.8742
	ordinal CE	0.929 ± 0.006	0.857 ± 0.013	0.8642
	CORAL	0.901 ± 0.007	0.827 ± 0.014	0.8534
	CORN	0.924 ± 0.006	0.857 ± 0.017	0.8683
	MSE	**0.930 ± 0.004**	**0.864 ± 0.011**	**0.8784**

Table B.3
Local H-ViT results when pretrained on TCGA dataset, for different region sizes and loss functions. We report the mean and standard deviation of the quadratic weighted kappa across the 5 cross-validation folds, along with the kappa score achieved by ensembling predictions from each fold.

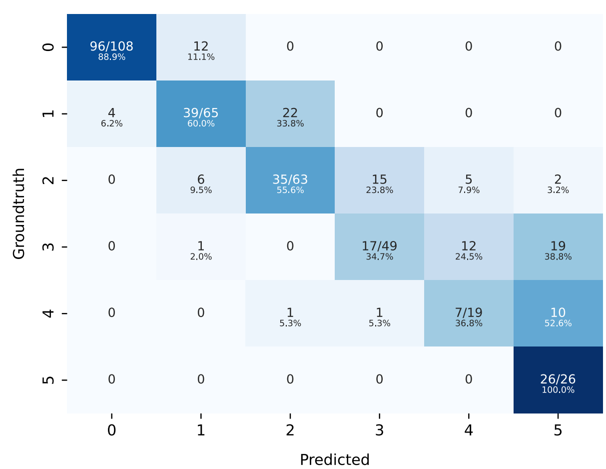
Region size	Loss	Tune	Test (public)	Test (public, *ens.*)
4096	CE	0.803 ± 0.017	0.761 ± 0.018	0.7732
	ordinal CE	0.816 ± 0.019	0.764 ± 0.021	0.7763
	CORAL	0.794 ± 0.014	0.757 ± 0.018	**0.7850**
	CORN	0.804 ± 0.017	0.765 ± 0.008	0.7844
	MSE	**0.826 ± 0.025**	**0.770 ± 0.020**	0.7818
2048	CE	0.814 ± 0.014	0.777 ± 0.023	0.8044
	ordinal CE	0.829 ± 0.020	0.788 ± 0.015	0.8055
	CORAL	0.816 ± 0.009	0.780 ± 0.010	0.8007
	CORN	**0.838 ± 0.016**	**0.796 ± 0.023**	**0.8080**
	MSE	0.832 ± 0.012	0.790 ± 0.013	0.8068
1024	CE	0.814 ± 0.011	0.774 ± 0.012	0.7926
	ordinal CE	**0.836 ± 0.009**	0.767 ± 0.030	0.7828
	CORAL	0.813 ± 0.015	0.752 ± 0.010	0.7771
	CORN	0.810 ± 0.043	**0.775 ± 0.025**	**0.7935**
	MSE	0.833 ± 0.012	0.773 ± 0.016	0.7859

Table B.4
Local H-ViT results when pretrained on PANDA dataset, for different region sizes and loss functions. We report the mean and standard deviation of the quadratic weighted kappa across the 5 cross-validation folds, along with the kappa score achieved by ensembling predictions from each fold.

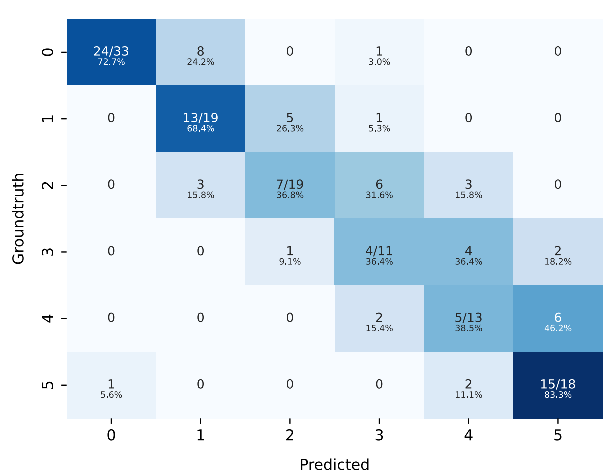
Region size	Loss	Tune	Test (public)	Test (public, *ens.*)
4096	CE	0.936 ± 0.008	0.882 ± 0.016	0.8809
	ordinal CE	0.939 ± 0.008	0.886 ± 0.017	0.8970
	CORAL	0.888 ± 0.008	0.852 ± 0.014	0.8643
	CORN	0.932 ± 0.009	0.883 ± 0.016	0.8998
	MSE	**0.940 ± 0.006**	**0.894 ± 0.011**	**0.9061**
2048	CE	0.942 ± 0.010	0.882 ± 0.011	0.8976
	ordinal CE	0.948 ± 0.004	0.894 ± 0.012	0.9053
	CORAL	0.896 ± 0.006	0.859 ± 0.008	0.8782
	CORN	0.941 ± 0.007	0.901 ± 0.011	0.9077
	MSE	**0.950 ± 0.002**	**0.904 ± 0.011**	**0.9149**
1024	CE	0.943 ± 0.005	0.878 ± 0.024	0.8911
	ordinal CE	0.946 ± 0.004	0.884 ± 0.013	0.8909
	CORAL	0.899 ± 0.006	0.868 ± 0.009	0.8787
	CORN	0.945 ± 0.005	0.893 ± 0.012	0.8977
	MSE	**0.949 ± 0.007**	**0.896 ± 0.009**	**0.9041**



(a) PANDA combined test set



(b) Karolinska University Hospital dataset



(c) Crowd-sourced dataset

Fig. C.1. Confusion matrices of our best performing ensemble model on the 3 evaluation datasets.

Appendix J. Computational requirements

See Tables L.11 and L.12.

Appendix K. Model ensembling

In a 5-fold cross-validation setup, each fold yields a trained model, resulting in five models in total. For inference on a given test set, each slide is passed through all five models, and their predictions are

Table D.5
Classification performance of our best ensemble Local H-ViT models against that of PANDA consortium teams on PANDA public and private test sets, as well as Karolinska University Hospital dataset, used as external validation data after the challenge ended. All values are given as overall accuracy.

Model	Test (public)	Test (private)	Test (public+private)	Karolinska University Hospital dataset
Save The Prostate	0.684	0.732	0.712	0.618
NS Pathology	**0.730**	**<u>0.750</u>**	**<u>0.742</u>**	**0.724**
PND	0.707	0.745	0.729	0.697
iafoss	0.712	0.721	0.717	0.521
Aksell	**<u>0.735</u>**	0.738	**0.737**	0.648
vanda	0.702	0.739	0.724	0.612
ChienYiChi	0.679	0.727	0.707	0.694
BarelyBears	0.707	0.749	0.731	0.706
rahma.ai	0.715	0.719	0.717	0.579
ctrasd123	0.705	**<u>0.750</u>**	0.731	**<u>0.736</u>**
Kiminya	0.695	0.745	0.724	0.670
Manuel Campos	0.707	0.727	0.719	0.709
Dmitry A. Grechka	0.672	0.721	0.700	0.406
KovaLOVE v2	0.674	0.714	0.697	0.691
Local H-ViT[a] (*ens.*)	0.702	0.728	0.717	0.667
Local H-ViT[b] (*ens.*)	0.700	0.719	0.711	0.673

[a] Starting from features given by our PANDA-pretrained ViT-S.
[b] Starting from features given by UNI.

Table D.6
Classification performance of our best ensemble Local H-ViT models against that of PANDA consortium teams on PANDA public and private test sets, as well as Karolinska University Hospital dataset, used as external validation data after the challenge ended. All values are given as binary accuracy for distinguishing between low-risk (ISUP $\leq$ 1) and higher-risk cases.

Model	Test (public)	Test (private)	Test (public+private)	Karolinska University Hospital dataset
Save The Prostate	0.913	0.930	0.923	0.898
NS Pathology	0.916	0.932	0.925	0.921
PND	0.913	0.930	0.923	0.906
iafoss	**0.924**	**<u>0.934</u>**	**<u>0.930</u>**	0.879
Aksell	**<u>0.926</u>**	0.910	0.917	0.915
vanda	0.921	**<u>0.934</u>**	**0.929**	0.909
ChienYiChi	0.913	0.928	0.922	**<u>0.933</u>**
BarelyBears	0.916	0.923	0.920	**0.930**
rahma.ai	0.911	0.916	0.914	0.903
ctrasd123	0.911	0.930	0.922	0.912
Kiminya	0.903	0.919	0.913	0.903
Manuel Campos	0.916	0.928	0.923	0.907
Dmitry A. Grechka	0.891	**<u>0.934</u>**	0.916	0.813
KovaLOVE v2	0.903	0.916	0.910	0.912
Local H-ViT[a] (*ens.*)	0.906	0.912	0.909	0.912
Local H-ViT[b] (*ens.*)	0.903	0.916	0.910	0.918

[a] Starting from features given by our PANDA-pretrained ViT-S.
[b] Starting from features given by UNI.

Table E.7
Average number of regions per slide, PANDA dataset.

Region size	M_{PANDA}
4096	6
2048	16
1024	46

ensembled via majority vote. When a tie occurs, two scenarios can exist:

1. each model predicts a distinct class
2. there is a tie between two classes, with a third class as an outlier

In the first scenario, we randomly select a class. In the second scenario, we opt for the class *closest* to the outlier. Instead of measuring the distance between two classes via absolute difference, we employed a slightly different distance function. This function penalizes transitions between benign (grade 0) and cancerous (grade 1 and above) predictions more severely than shifts between consecutive ISUP grades:

$$d(c_{\text{tied}}, c_{\text{outlier}}) = \begin{cases} 1.5 \text{ if } (c_{\text{tied}}, c_{\text{outlier}}) = (0,1) \text{ or if } (c_{\text{tied}}, c_{\text{outlier}}) = (1,0) \\ |c_{\text{tied}} - c_{\text{outlier}}| \text{ otherwise} \end{cases}$$

Appendix L. Position embedding interpolation

Hierarchical pretraining is not region-size independent as the dimension of the positional embedding layer depends on the number of input tokens. When working with smaller regions, we can either interpolate the position embedding learned on (4096,4096) regions to match the smaller input sequence length, or learn a new one by pretraining a different region-level Transformer on the smaller regions.

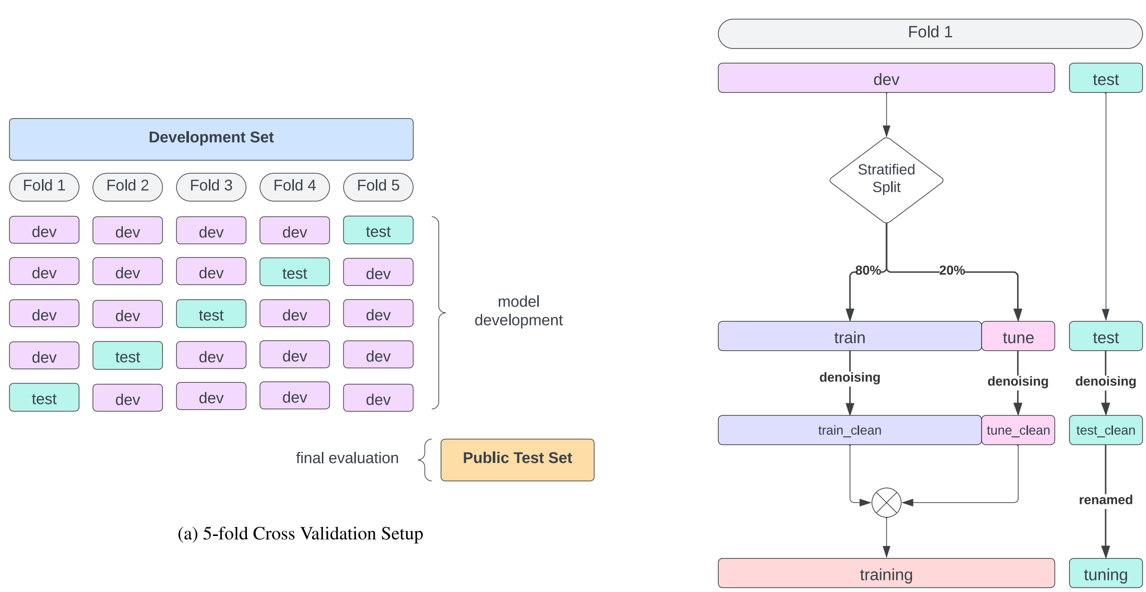
We experimented with both options when working with (2048, 2048) regions. The results, summarized in Table J.10, indicate that the latter approach generalizes better to PANDA public test set.

Table F.8
Test datasets details.

	PANDA public set	PANDA private set	Karolinska University Hospital dataset	Crowd-sourced dataset
Sources	Radboud University Medical Center (Netherlands) Karolinska Institutet (Sweden)	Radboud University Medical Center (Netherlands) Karolinska Institutet (Sweden)	Karolinska University Hospital (Sweden)	Rennes University Hospital (France) Radboud University Medical Center (Netherlands) Institute of Pathology and Molecular Immunology of the University of Porto (Portugal) Memorial Hospitals Istanbul (Turkey) Foch Hospital (France) Isala Zwolle (Netherlands)
Scanning Equipments	3DHistech Pannoramic Flash II 250 Aperio AT2 Hamamatsu C9600-12	3DHistech Pannoramic Flash II 250 Aperio AT2 Hamamatsu C9600-12	Hamamatsu C13220-01	3DHistech P1000 Hamamatsu Aperio AT2 Phillips KFBIO
Number of Samples	393	545	330	113

Table G.9
Number of slide per partition after label denoising, 5-fold CV splits.

Fold	Training set (cleaned)	Tuning set (cleaned)
0	7620	1933
1	7633	1920
2	7636	1917
3	7653	1900
4	7670	1883



(a) 5-fold Cross Validation Setup

(b) Fold Splitting Strategy

Fig. G.2. Overview of the 5-fold Cross Validation Splitting Strategy.

(a) `block_id = 0`

(b) `block_id = 122`

(c) `block_id = 3`

(d) `block_id = 2`

Fig. H.3. Visualization of four artificial blocks. **(a)**, **(c)** and **(d)** show blocks with 2 slides. **(b)** shows a block with 3 slides.

Table J.10
Global H-ViT results when pretrained on PANDA dataset with (2048, 2048) regions, for different loss functions. We report quadratic weighted kappa averaged over the 5 cross-validation folds (mean ± std).

Pretraining region size	Input region size	Loss	Tune	Test (public)
4096	2048	CE	0.862 ± 0.031	0.752 ± 0.050
		ordinal CE	0.879 ± 0.022	0.774 ± 0.039
		CORAL	0.872 ± 0.011	0.748 ± 0.053
		CORN	0.888 ± 0.014	0.783 ± 0.050
		MSE	0.890 ± 0.014	0.784 ± 0.052
2048	2048	CE	0.871 ± 0.018	**0.798 ± 0.035**
		ordinal CE	0.888 ± 0.022	**0.783 ± 0.028**
		CORAL	0.870 ± 0.017	**0.773 ± 0.021**
		CORN	0.880 ± 0.012	**0.790 ± 0.033**
		MSE	0.888 ± 0.019	**0.799 ± 0.018**

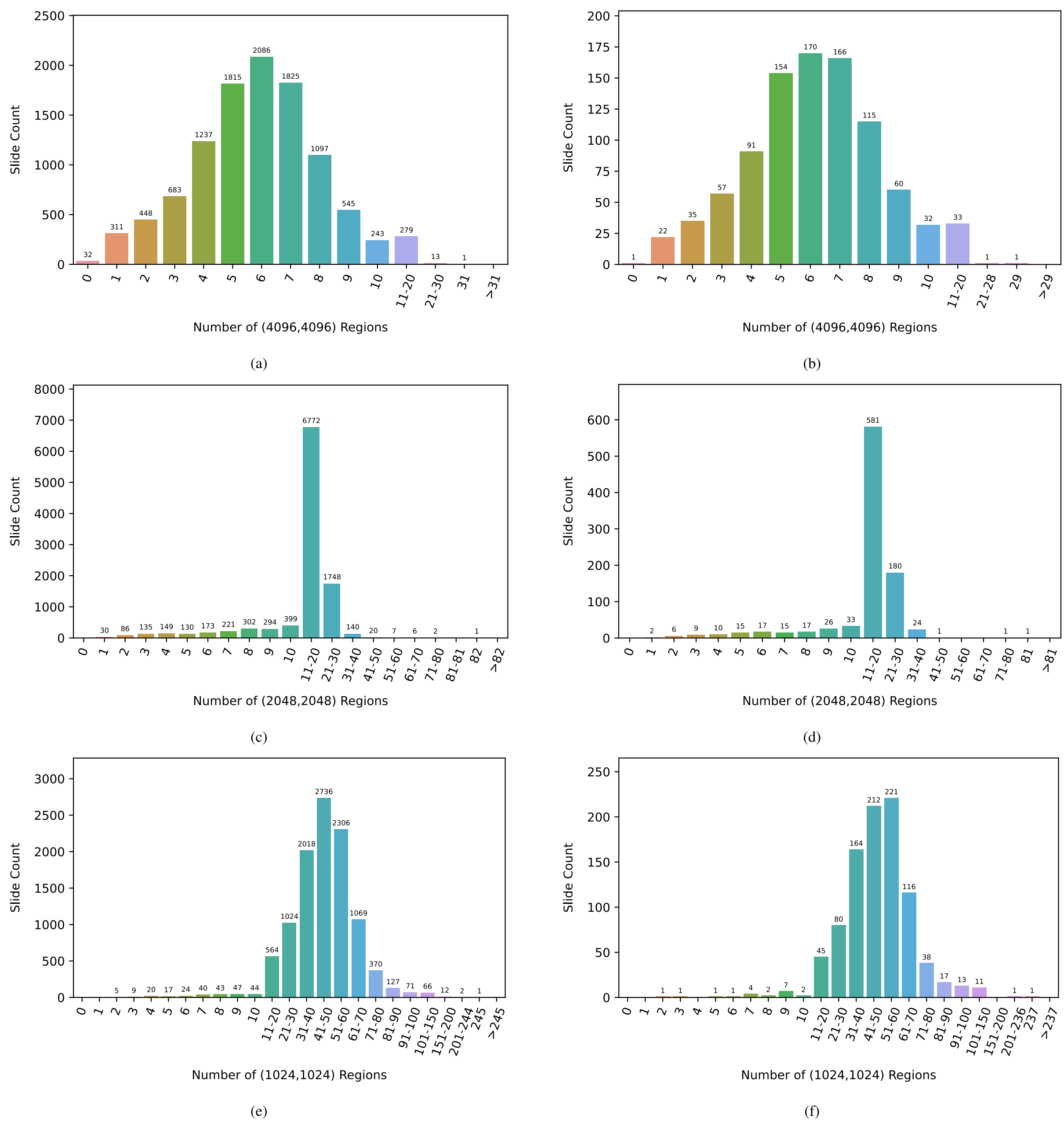


Fig. I.4. Distribution of the number of extracted regions for PANDA development set with **(a)** (4096, 4096) regions **(c)** (2048, 2048) regions **(e)** (1024, 1024) regions and for PANDA public test set with **(b)** (4096, 4096) regions **(d)** (2048, 2048) regions **(f)** (1024, 1024) regions.

Table L.11
Computational characteristics of DINO pretraining of the path-level and region-level Transformer on PANDA.

Transformer	Architecture	# Parameters	# Epochs	Input size	Pretraining time	GPU card
patch-level	ViT-Small	22 M	50	$(256, 256)$	72h	GeForce RTX 3080 Ti
region-level	ViT-Tiny	5.7 M	125	$(4096, 4096)$	7h	GeForce RTX 3080 Ti
region-level	ViT-Tiny	5.7 M	125	$(2048, 2048)$	17h	GeForce RTX 3080 Ti
region-level	ViT-Tiny	5.7 M	125	$(1024, 1024)$	50h	GeForce RTX 3080 Ti

Table L.12
Comparison of number of trainable parameters, training time, and GPU memory usage for Global H-ViT and Local H-ViT.

Model	# Parameters	Training time	GPU memory used	GPU card
Global H-ViT	0.6 M	1min 8s/epoch	1.1 GB	GeForce RTX 2080 Ti
Local H-ViT	3.5 M	5min 30s/epoch	1.7 GB	GeForce RTX 2080 Ti

Data availability

The data and code are publicly available. Data: https://www.kaggle.com/c/prostate-cancer-grade-assessment/data Code: https://github.com/computationalpathologygroup/hvit.

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